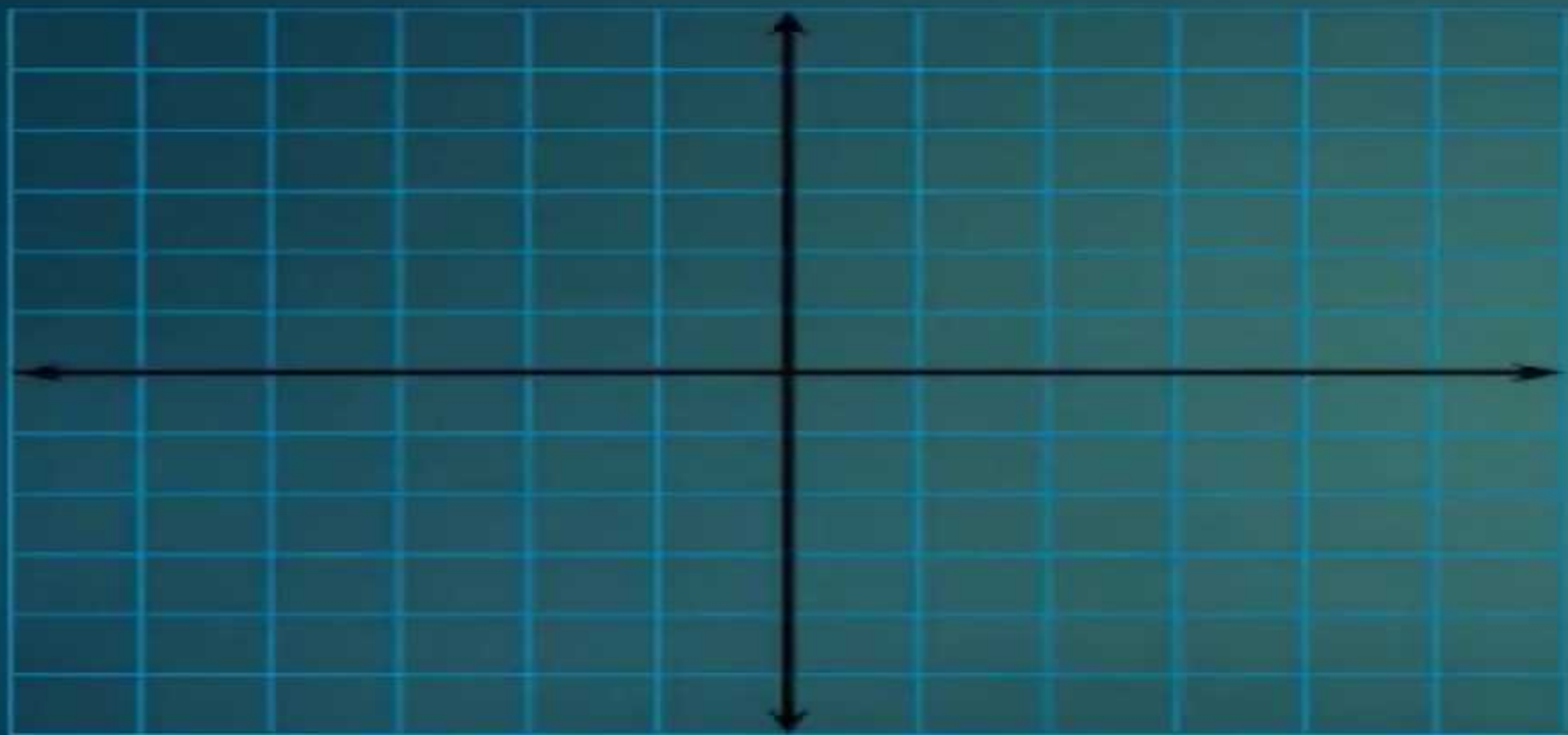
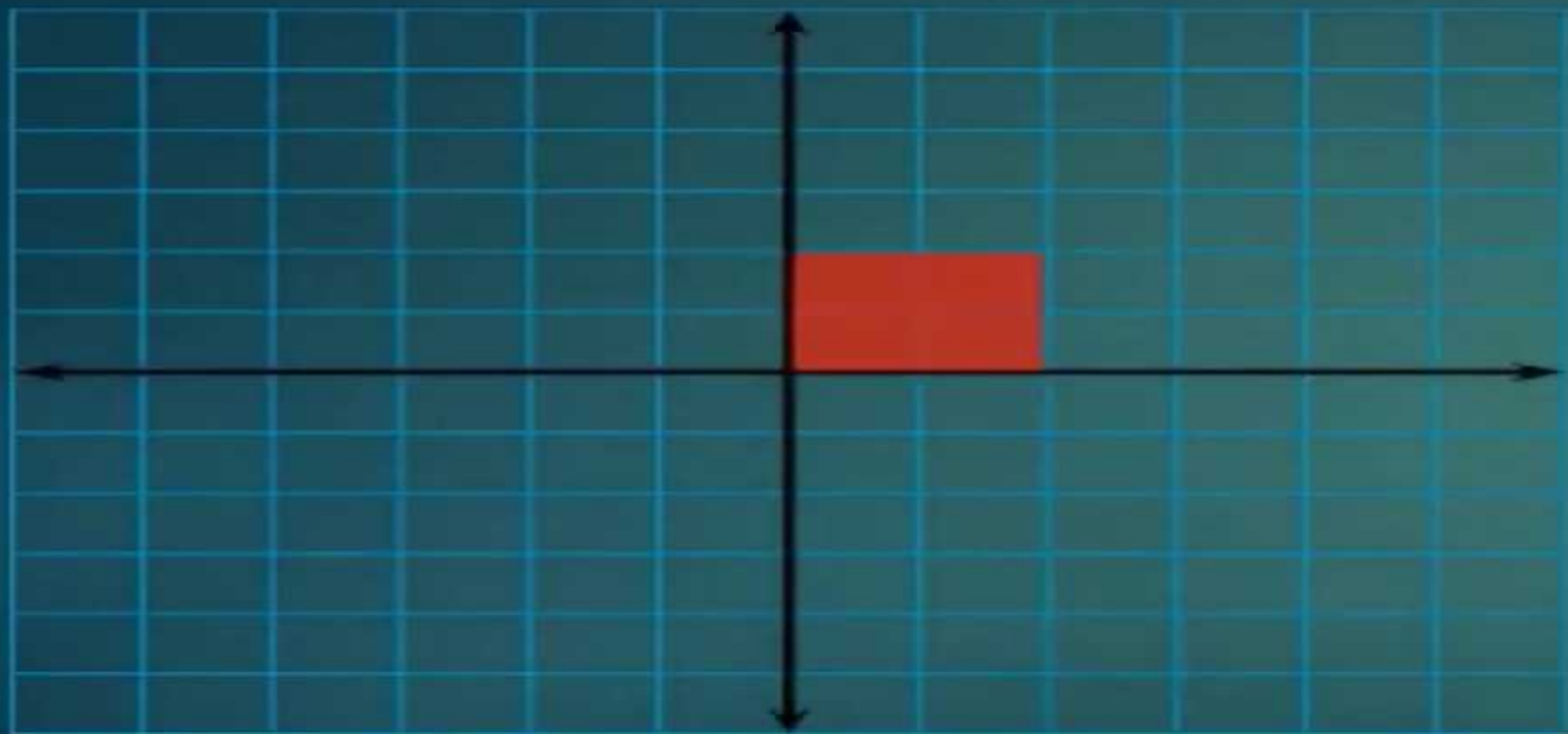
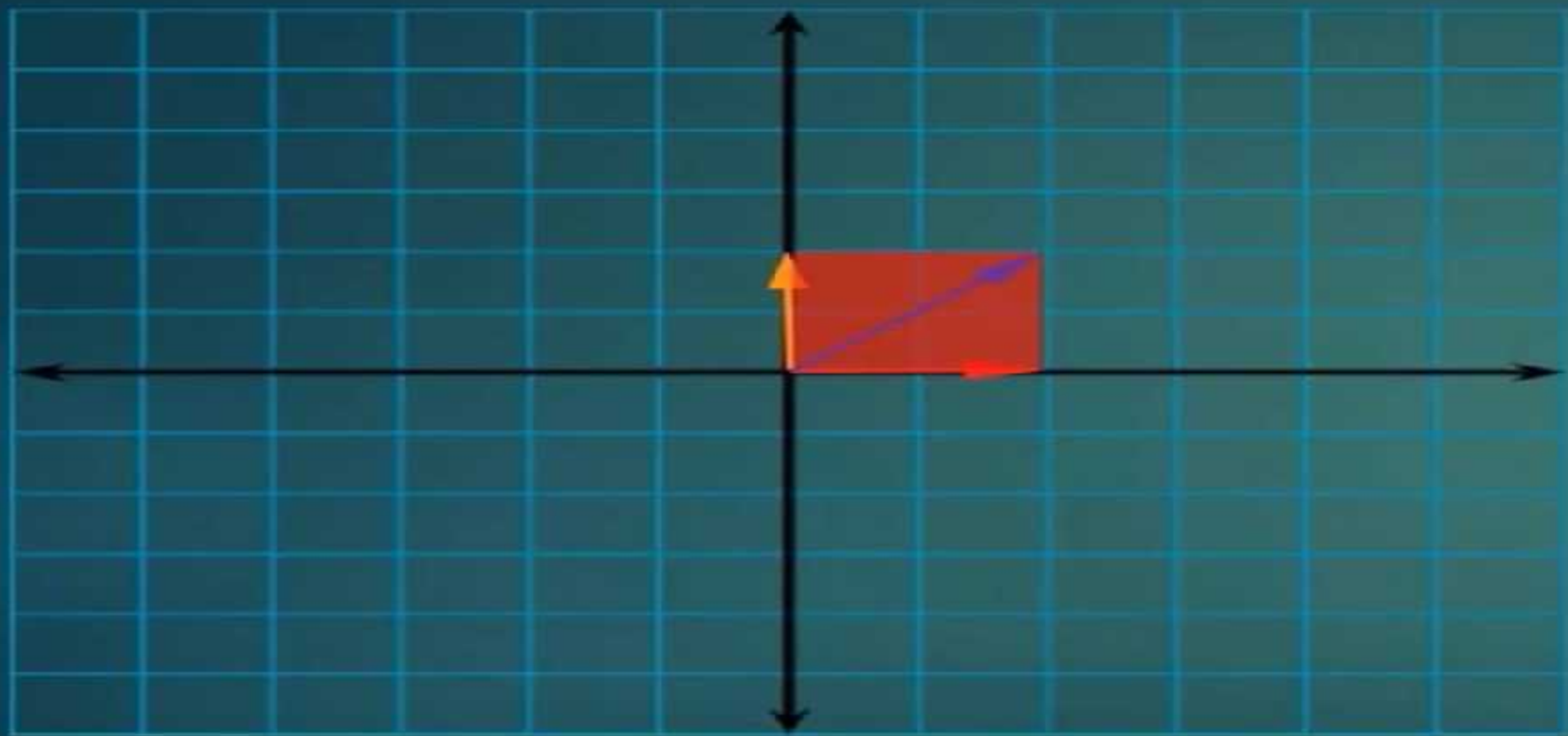
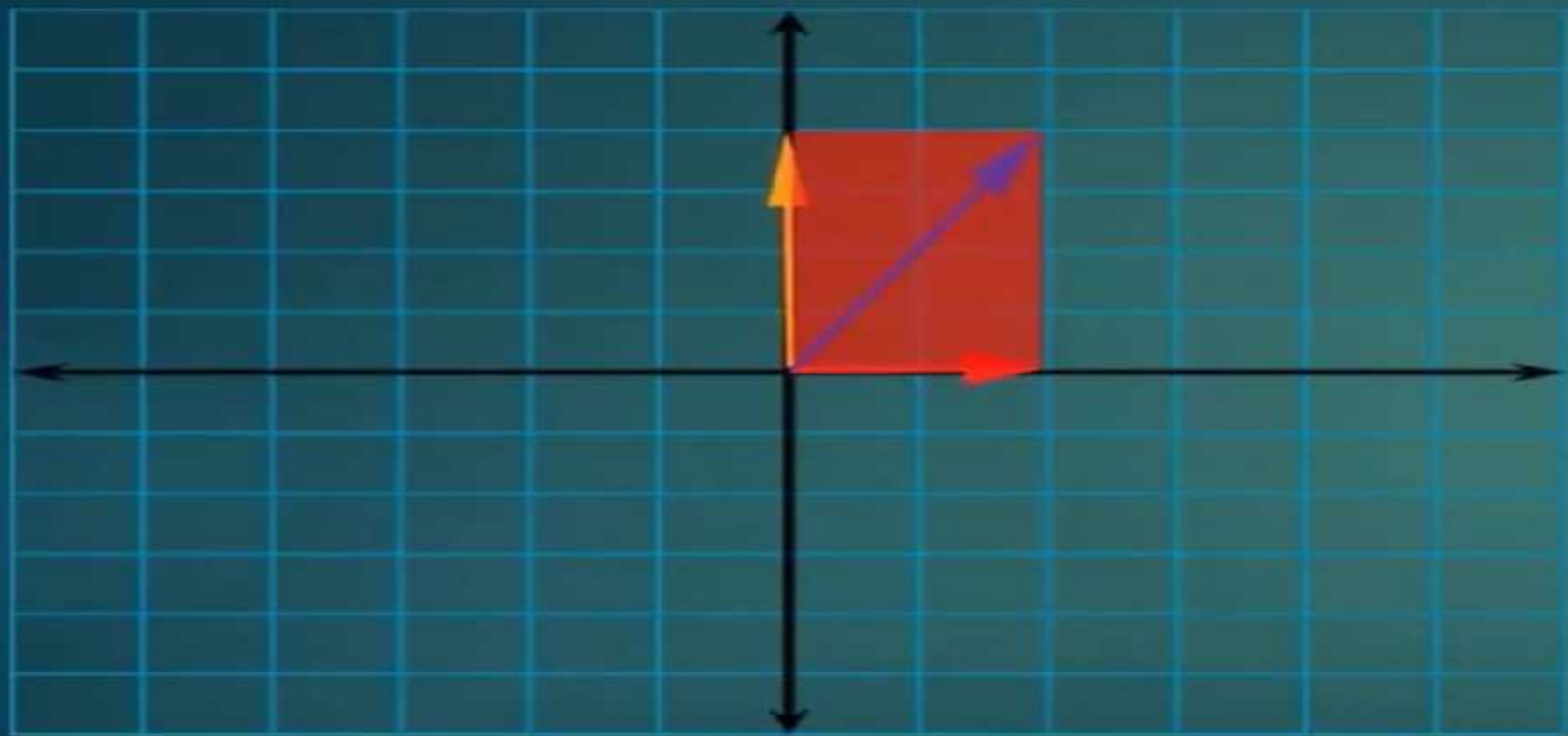
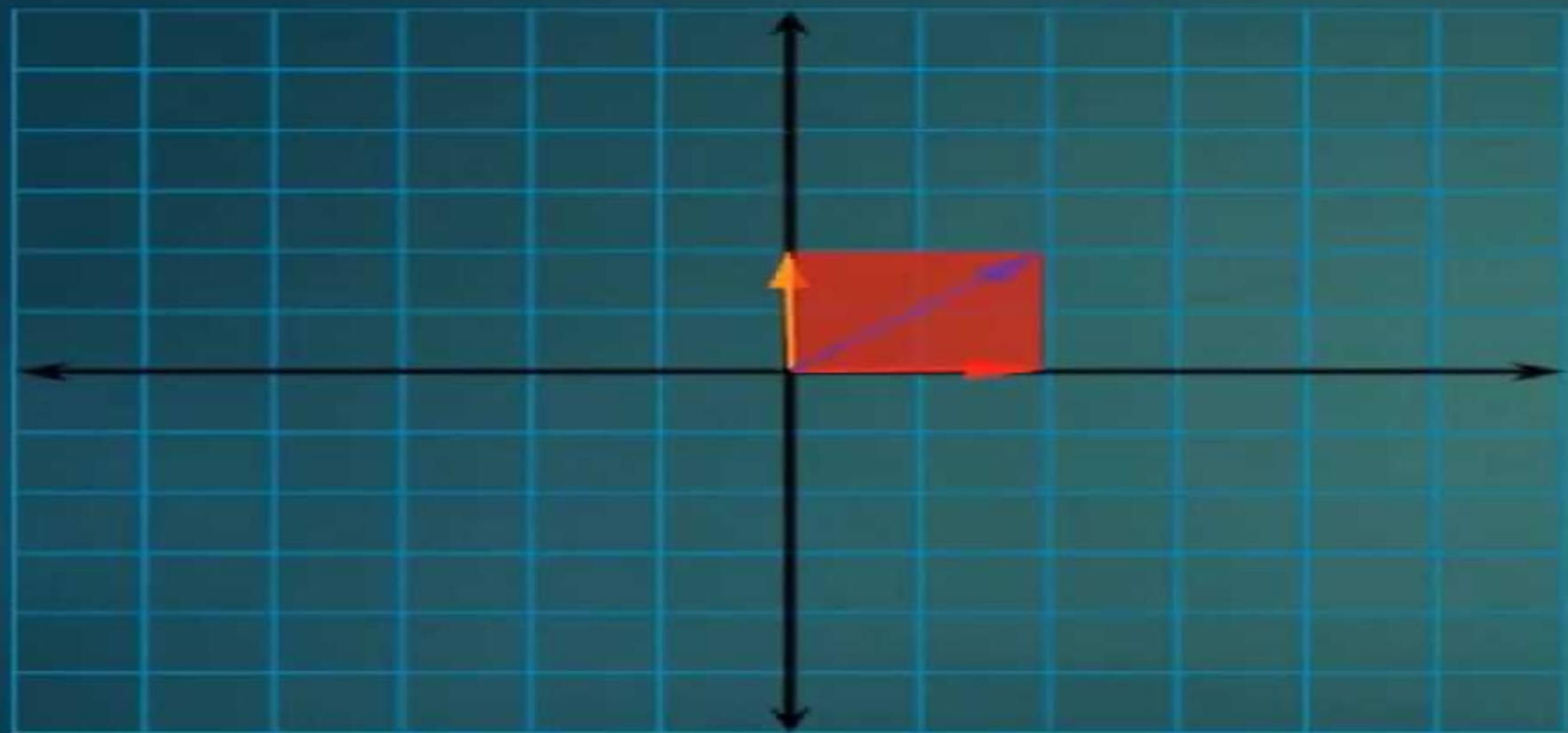


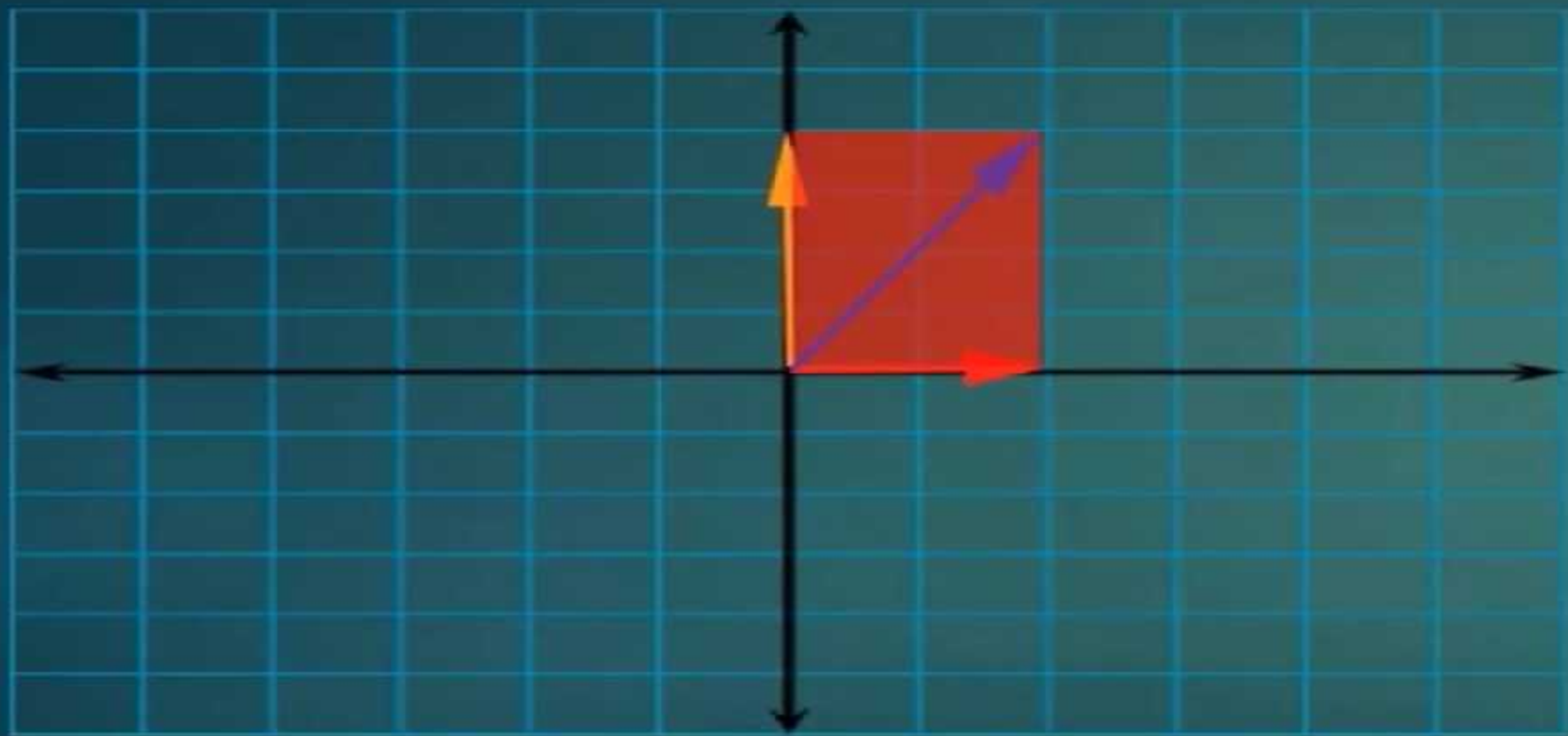


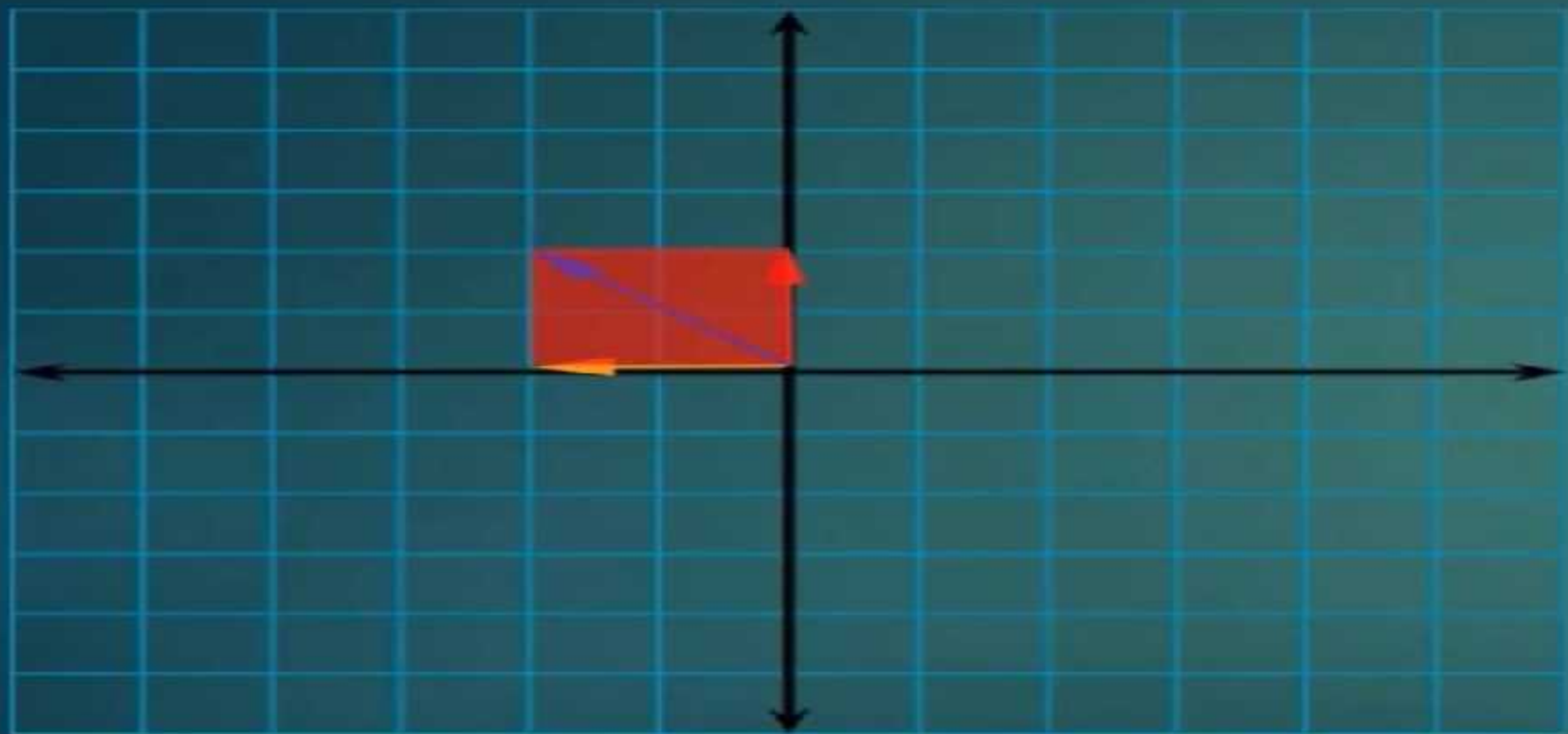


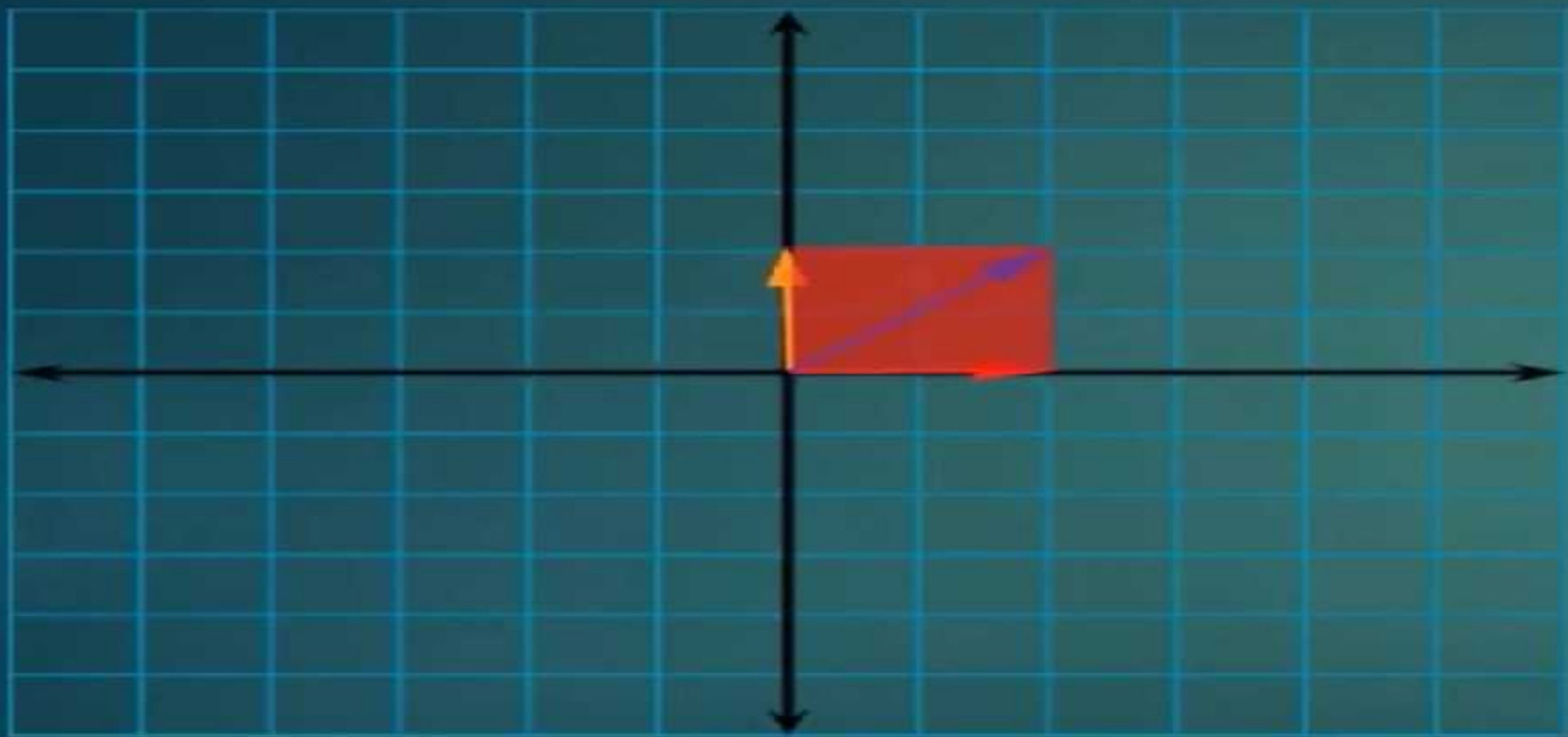


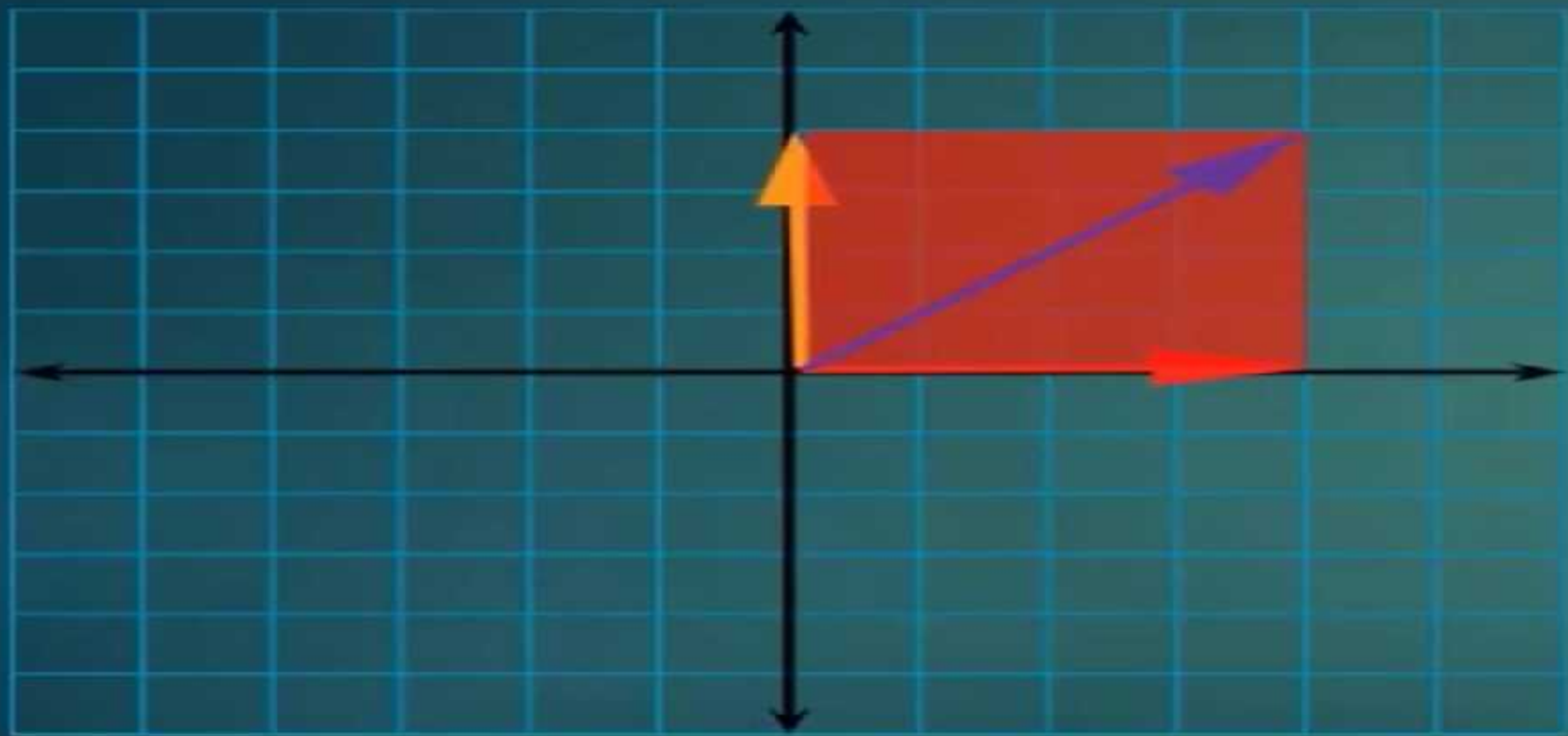


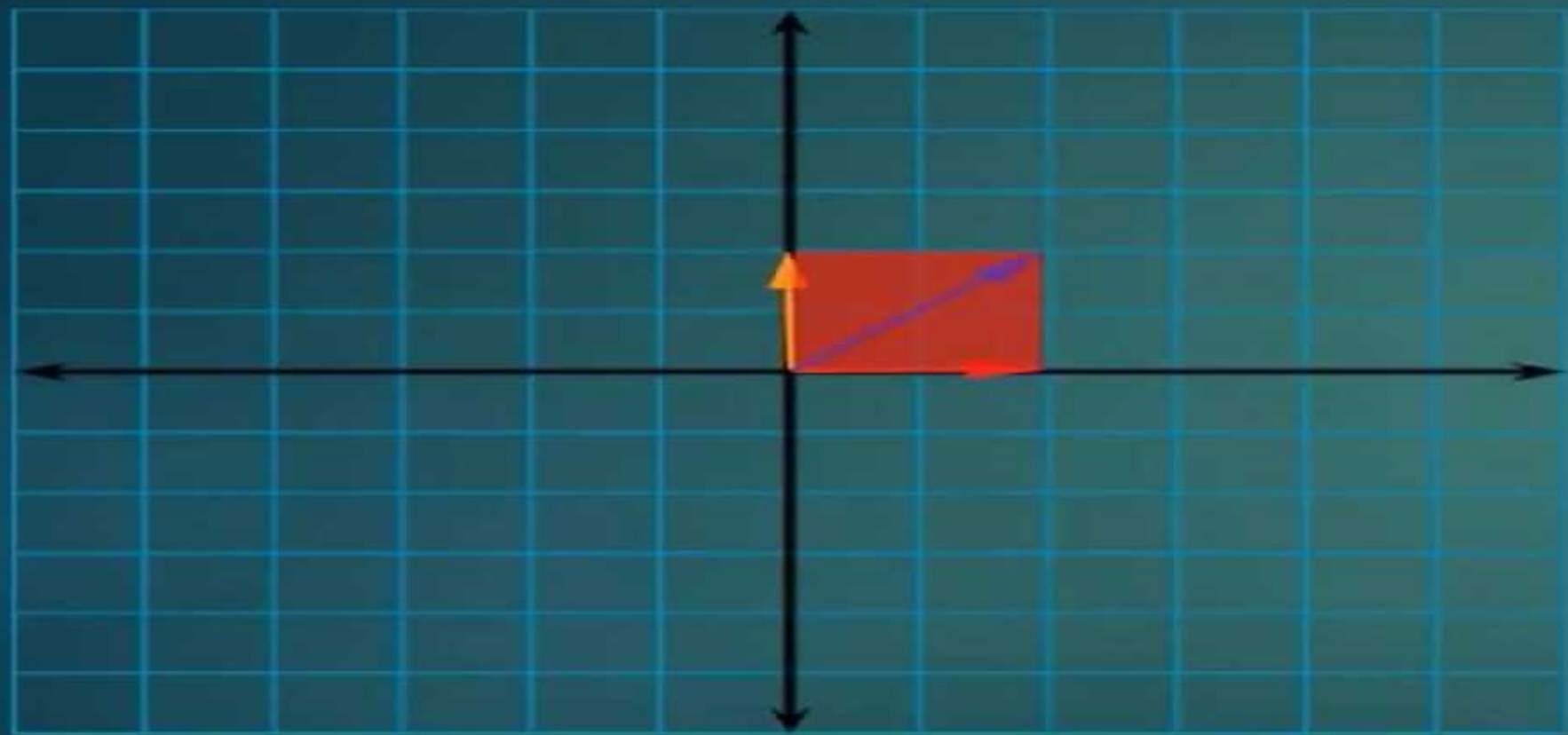


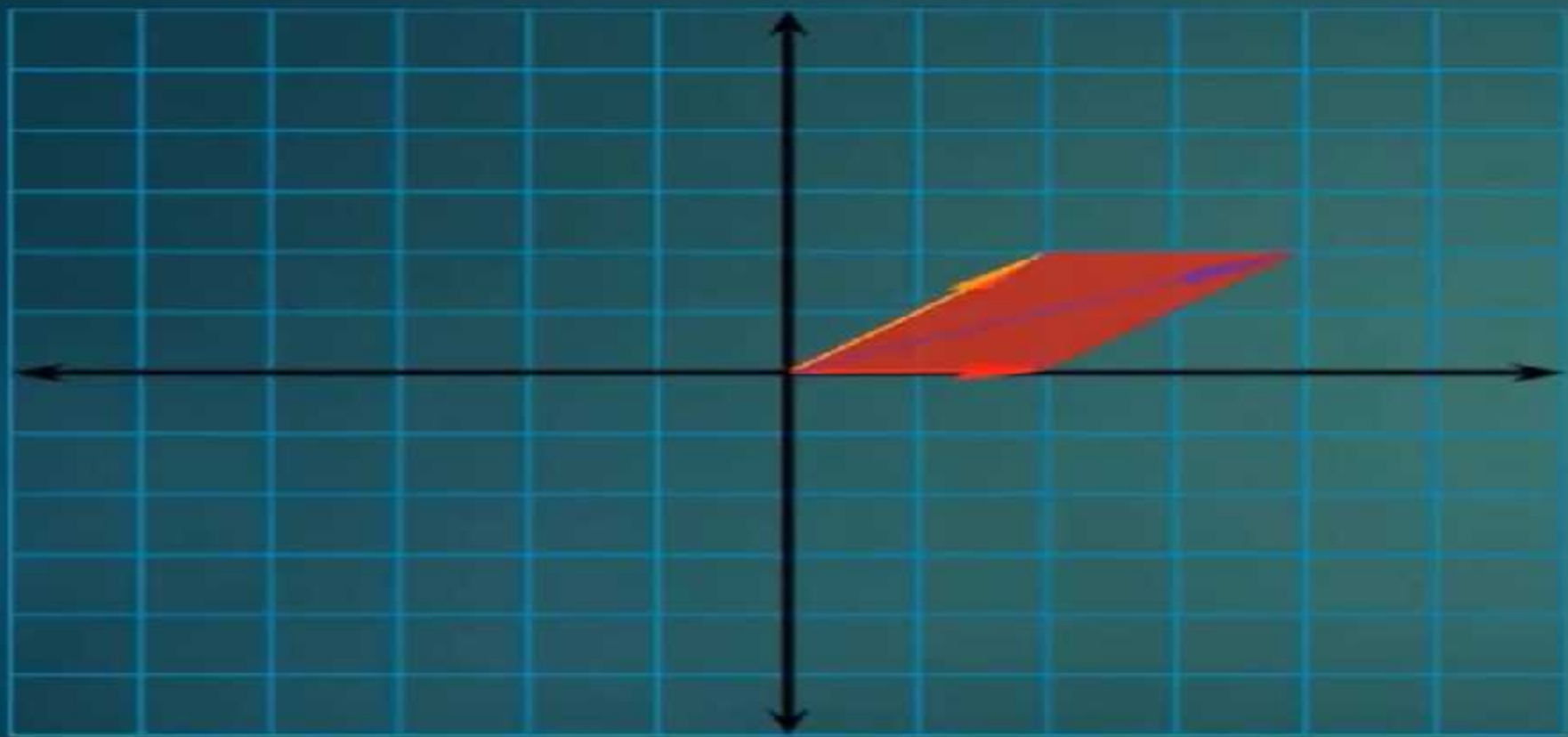


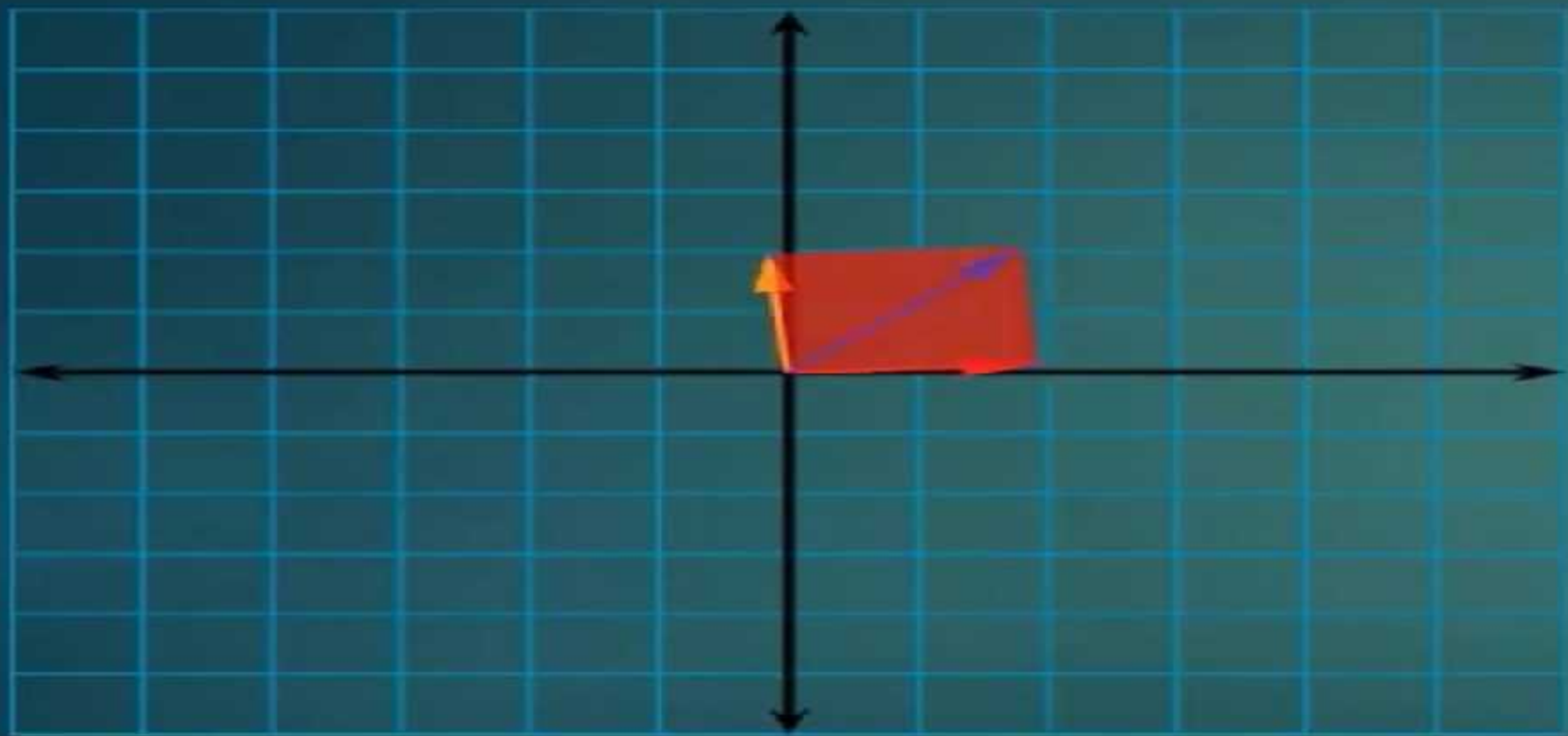


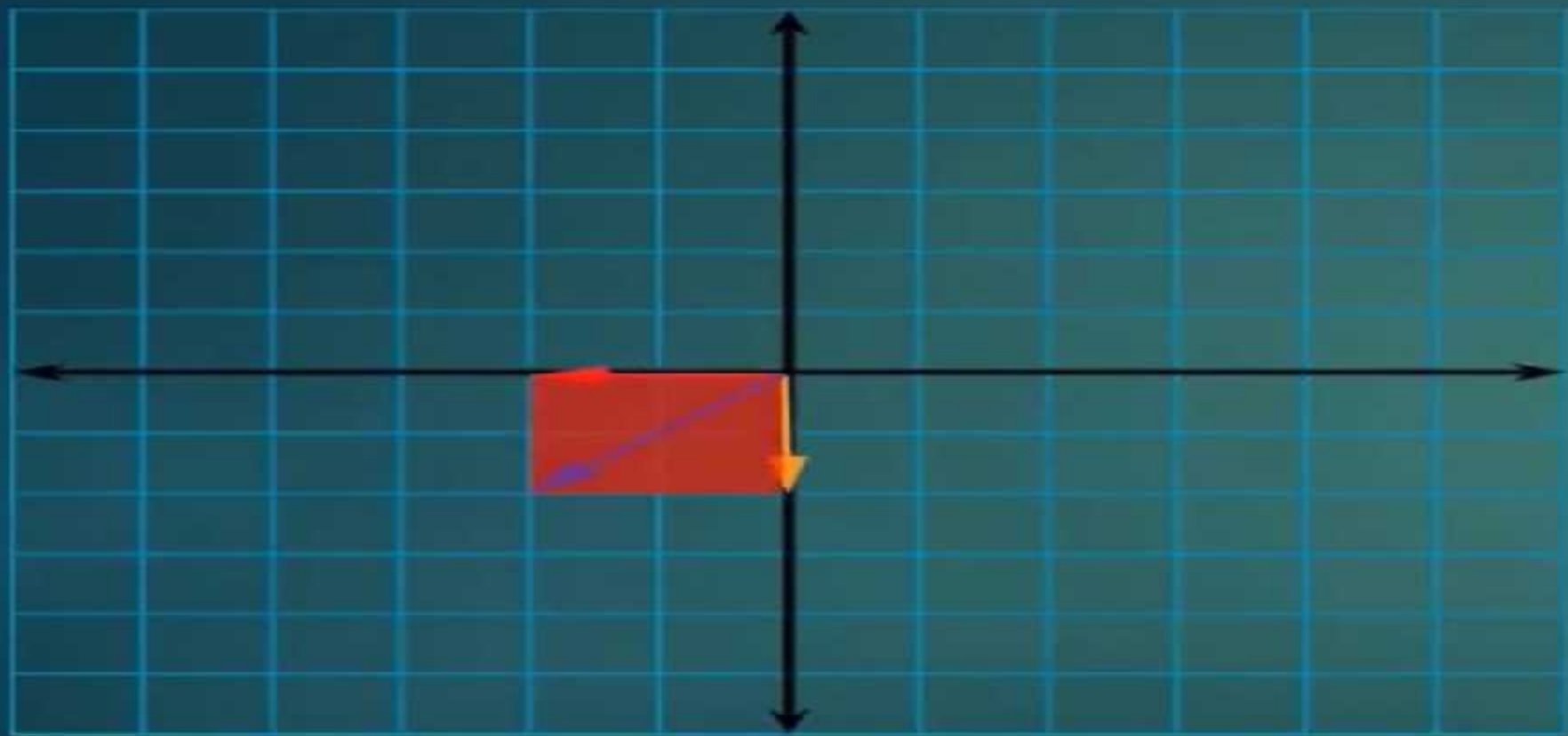












$$Av = \lambda v$$

$$Av = \lambda v$$

$$Av - \lambda v = 0$$

$$Av = \lambda v$$

$$Av - \lambda v = 0$$

$$(A - \lambda)v = 0$$

$$Av = \lambda v$$

$$Av - \lambda v = 0$$

$$(A - \lambda)v = 0$$

$$(A - \lambda I)v = 0$$

$$|A - \lambda I| = 0$$

$$|A - \lambda I| = 0$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$(A - \lambda I) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$(A - \lambda I) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

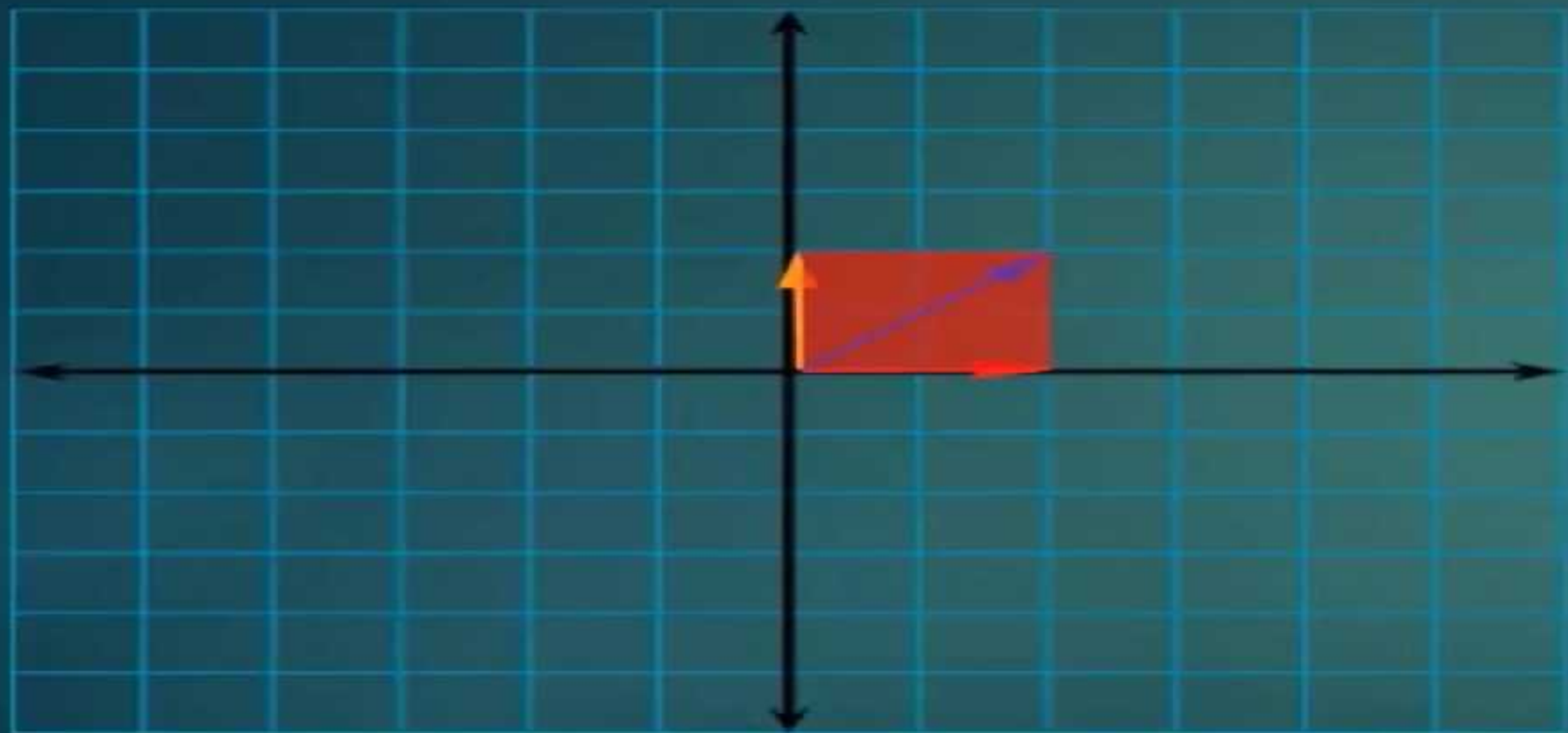
$$= \begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

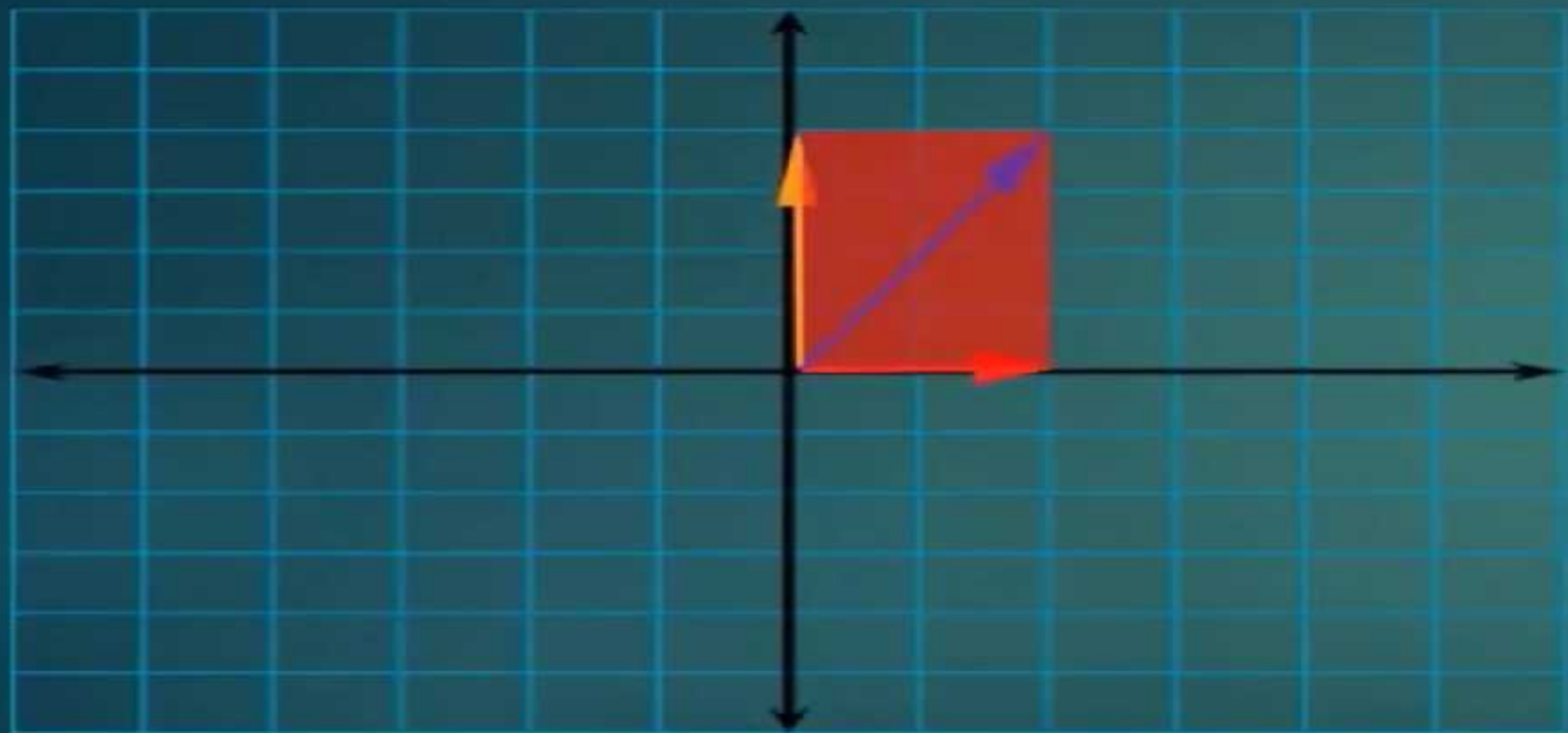
$$= \begin{bmatrix} a - \lambda & b \\ c & d - \lambda \end{bmatrix}$$

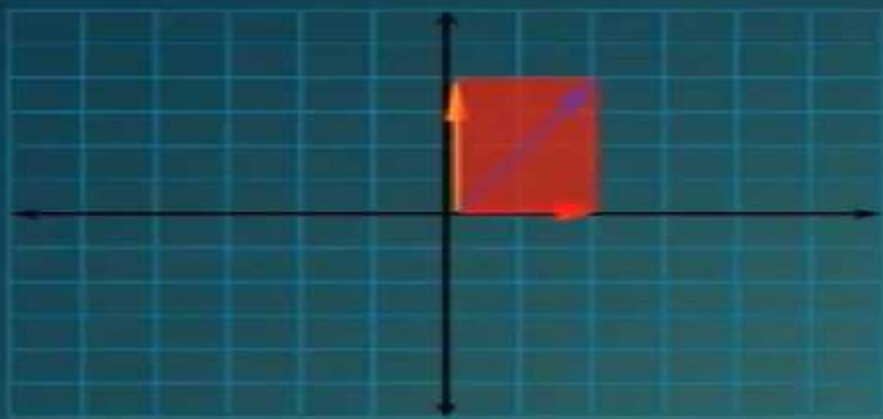
$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$= \begin{bmatrix} a - \lambda & b \\ c & d - \lambda \end{bmatrix}$$

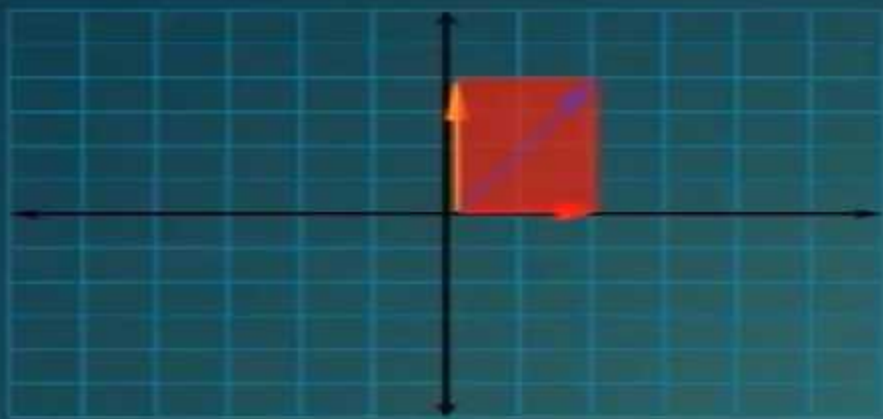
$$= \begin{bmatrix} a - \lambda & b \\ c & d - \lambda \end{bmatrix}$$







$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$



$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 0 \\ 0 & 2 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)(2 - \lambda) = 0$$

$$(1 - \lambda)(2 - \lambda) = 0$$

$$\lambda = 1 \text{ or } \lambda = 2$$

$$(1 - \lambda)(2 - \lambda) = 0$$

$$\lambda = 1 \text{ or } \lambda = 2$$

$$@\lambda = 1$$

$$(1 - \lambda)(2 - \lambda) = 0$$

$$\lambda = 1 \text{ or } \lambda = 2$$

$$@\lambda = 1$$

$$(A - \lambda I)v = 0$$

$$(A - \lambda I)v = 0$$

$$\begin{bmatrix} 1 - \lambda & 0 \\ 0 & 2 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$(A - \lambda I)v = 0$$

$$\begin{bmatrix} 1 - \lambda & 0 \\ 0 & 2 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$(A - \lambda I)v = 0$$

$$\begin{bmatrix} 1 - \lambda & 0 \\ 0 & 2 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$0x + 0y = 0$$

$$(A - \lambda I)v = 0$$

$$\begin{bmatrix} 1 - \lambda & 0 \\ 0 & 2 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$0x + 0y = 0$$

$$0x + 1y = 0$$

$$(A - \lambda I)v = 0$$

$$\begin{bmatrix} 1 - \lambda & 0 \\ 0 & 2 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$0x + 0y = 0$$

$$0x + 1y = 0$$

$$y = 0$$

$$v = \begin{bmatrix} t \\ 0 \end{bmatrix}$$

$$@\lambda = 2$$

$$@\lambda = 2$$

$$(A - \lambda I)v = 0$$

$$@\lambda = 2$$

$$(A - \lambda I)v = 0$$

$$\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$@\lambda = 2$$

$$(A - \lambda I)v = 0$$

$$\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$-1x + 0y = 0$$

$$@\lambda = 2$$

$$(A - \lambda I)v = 0$$

$$\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

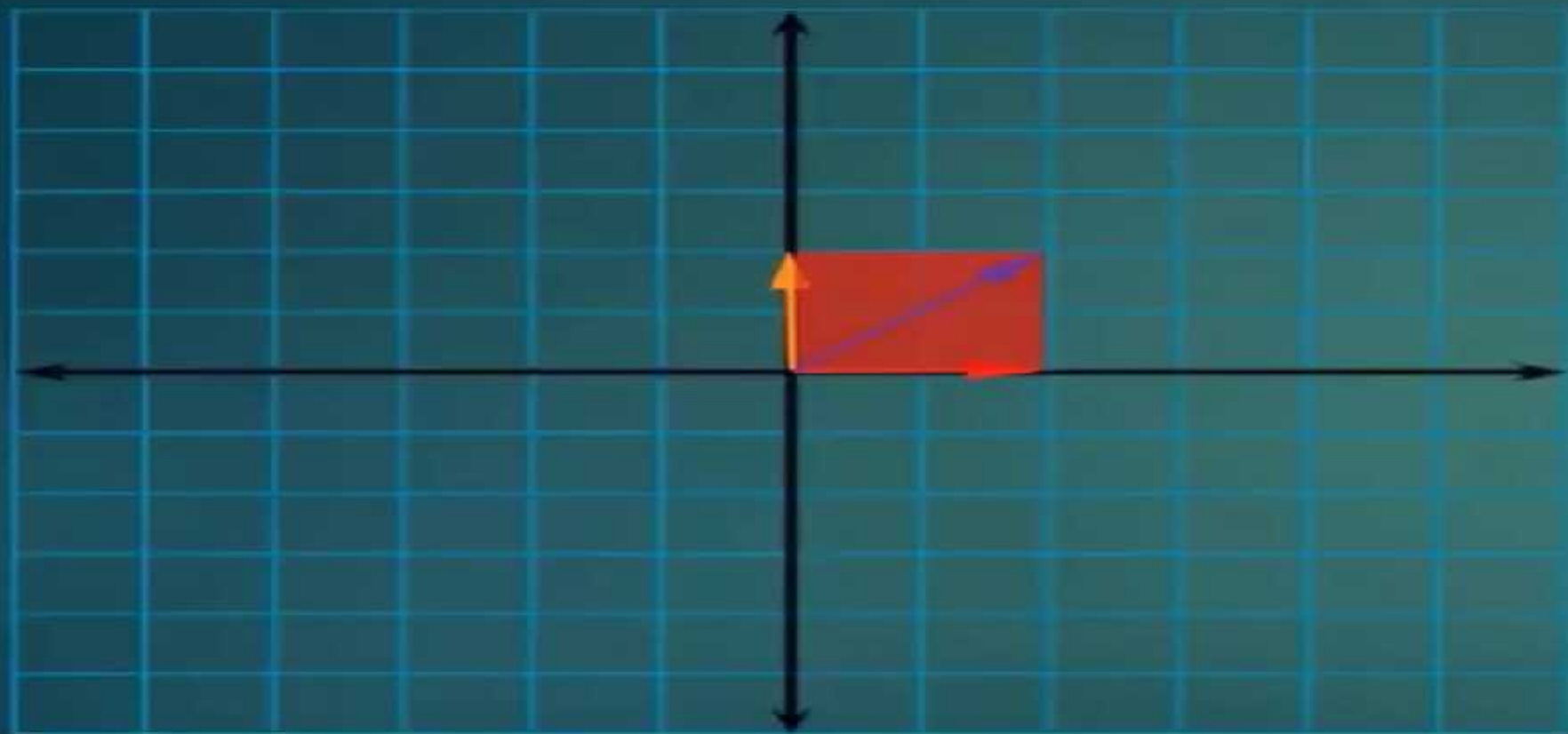
$$-1x + 0y = 0$$

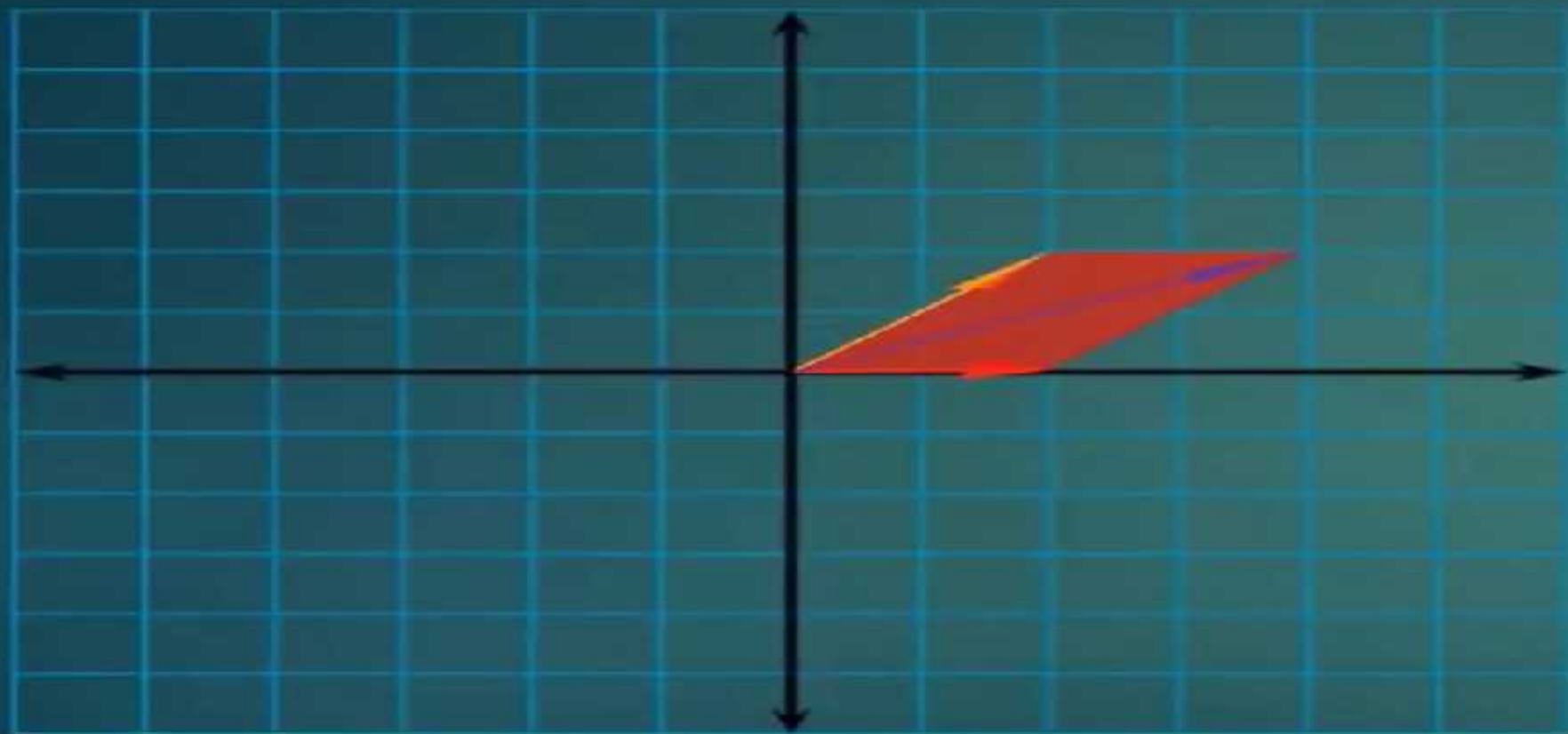
$$0x + 0y = 0$$

$$x = 0$$

$$v = \begin{bmatrix} 0 \\ t \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$





$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 0 \\ 0 & 1 - \lambda \end{vmatrix} = 0$$

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 0 \\ 0 & 1 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)(1 - \lambda) = 0$$

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 0 \\ 0 & 1 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)(1 - \lambda) = 0$$

$$\lambda = 1$$

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 0 \\ 0 & 1 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)(1 - \lambda) = 0$$

$$\lambda = 1$$

$$(A - \lambda I)v = 0$$

$$(A - \lambda I)v = 0$$

$$\begin{bmatrix} 1 - \lambda & 1 \\ 0 & 1 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$(A - \lambda I)v = 0$$

$$\begin{bmatrix} 1 - \lambda & 1 \\ 0 & 1 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$(A - \lambda I)v = 0$$

$$\begin{bmatrix} 1 - \lambda & 1 \\ 0 & 1 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$0x + y = 0$$

$$(A - \lambda I)v = 0$$

$$\begin{bmatrix} 1 - \lambda & 1 \\ 0 & 1 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

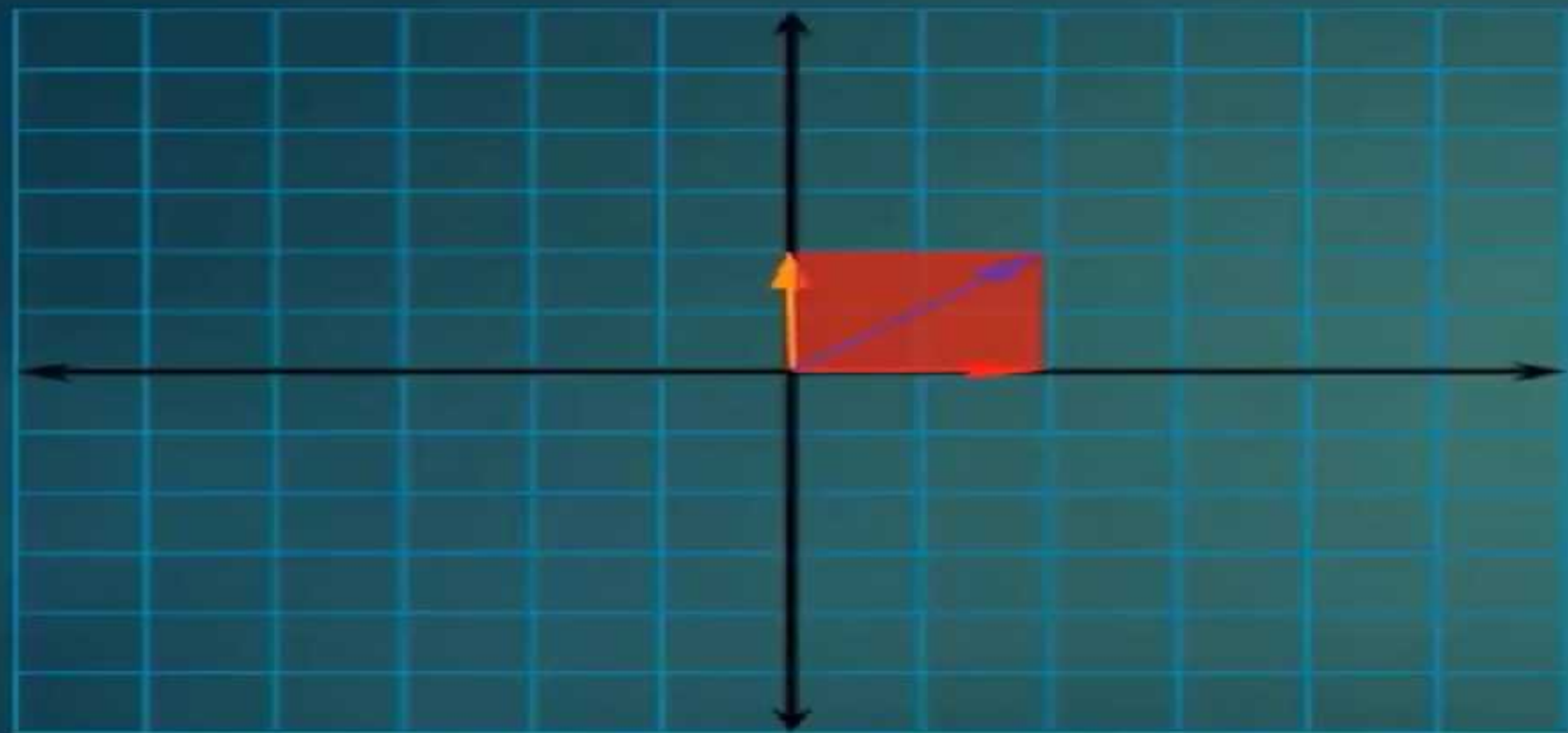
$$0x + y = 0$$

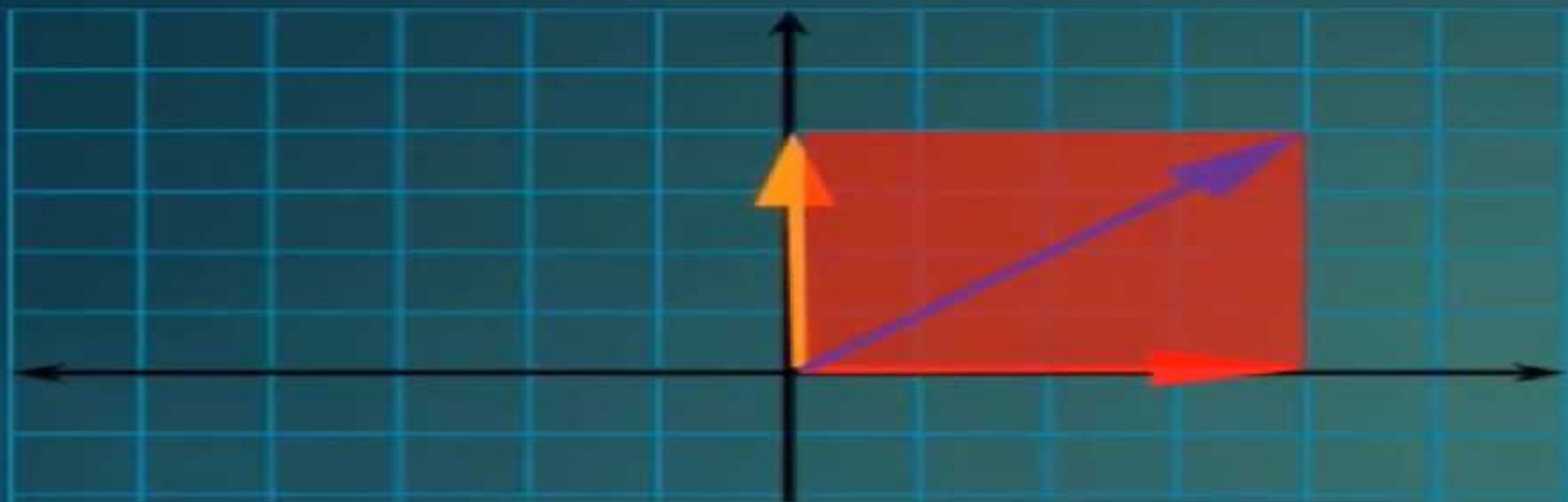
$$0x + 0y = 0$$

$$y = 0$$

$$y = 0$$

$$v = \begin{bmatrix} t \\ 0 \end{bmatrix}$$





$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & 0 \\ 0 & 2 - \lambda \end{vmatrix} = 0$$

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & 0 \\ 0 & 2 - \lambda \end{vmatrix} = 0$$

$$(2 - \lambda)(2 - \lambda) = 0$$

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & 0 \\ 0 & 2 - \lambda \end{vmatrix} = 0$$

$$(2 - \lambda)(2 - \lambda) = 0$$

$$\lambda = 2$$

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & 0 \\ 0 & 2 - \lambda \end{vmatrix} = 0$$

$$(2 - \lambda)(2 - \lambda) = 0$$

$$\lambda = 2$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & 0 \\ 0 & 2 - \lambda \end{vmatrix} = 0$$

$$(2 - \lambda)(2 - \lambda) = 0$$

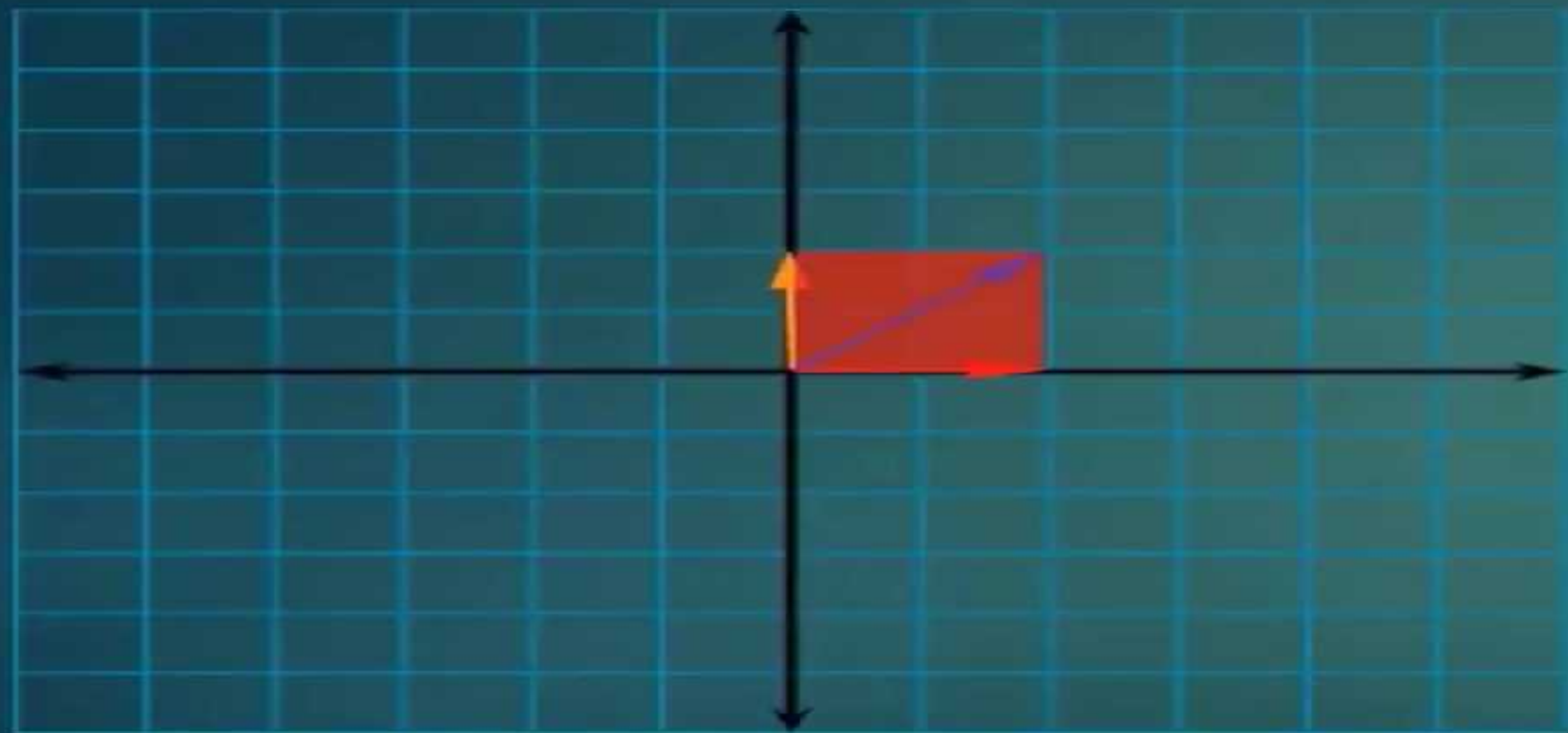
$$\lambda = 2$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

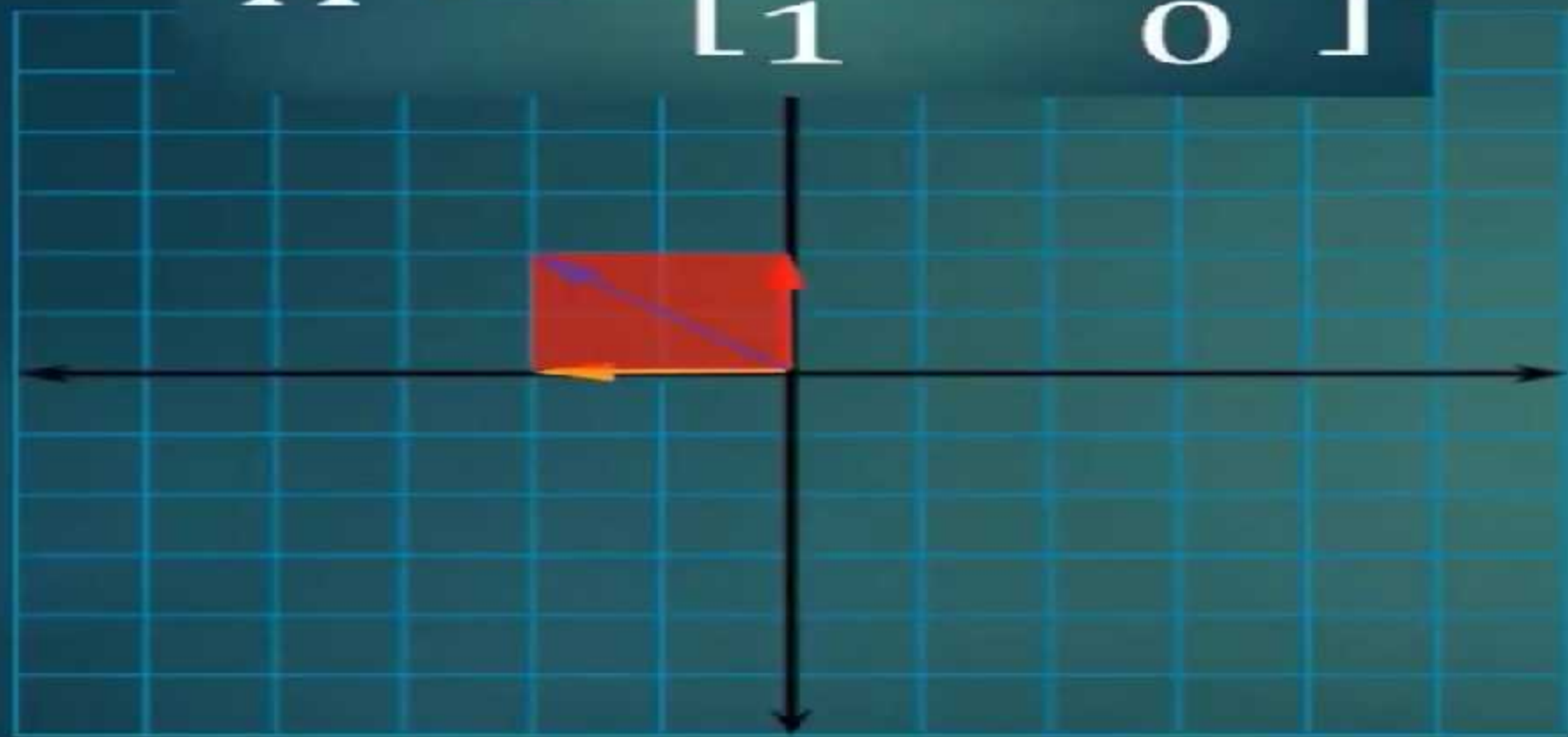
$$0x + 0y = 0$$

$$0x + 0y = 0$$

$$v = \begin{bmatrix} m \\ n \end{bmatrix}$$



$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$



$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$(-\lambda)(-\lambda) + 1 = 0$$

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$(-\lambda)(-\lambda) + 1 = 0$$

$$\lambda^2 + 1 = 0$$

$$\lambda^2 + 1 = 0$$

There is no Eigenvalue or
Eigenvector